

Oncology

Spontaneous regression of metastatic renal cancer and improvement in IMDC risk criteria prior to cytoreductive nephrectomy

Kay McGillivray^{*}, Roisin McKenna, Grenville Oades, Flora Rodger

Queen Elizabeth University Hospital Campus, UK

A B S T R A C T

A 57-year-old male presented with a 2-month history of a cough, weight loss, chest discomfort and night sweats. He was diagnosed with poor prognosis metastatic Renal Cancer (RCC) according to International Metastatic RCC Database Consortium (IMDC) criteria. We observed spontaneous regression of his metastatic disease and concurrent improvement in his IMDC risk stratification. This changed the initial recommendation of the MDT, and the patient elected to undergo cytoreductive nephrectomy (CRN) with the expectation he would need deferred systemic treatment. He made an uneventful post-operative recovery and at 12 months his follow-up imaging continues to show resolution of his metastatic disease.

1. Background

Renal Cancer is the 7th commonest Cancer in the UK accounting for 4% of all malignancies and up to one third of patients will have metastases at diagnosis. Since the turn of the millennium, we have seen huge strides in the management of patients with metastatic RCC. With the advent of therapies targeting the vascular endothelial growth factor (VEGF) pathway or the mammalian target of rapamycin pathway (MTOR) and more recently immune checkpoint inhibitors, the role of cytoreductive nephrectomy (CRN) prior to systemic treatment has been questioned. Current EAU guidelines only recommend up front CRN in patients with good performance status who do not require immediate systemic therapy.

Here we report a case of spontaneous regression of metastatic RCC and concurrent improvement in the risk stratification of a patient resulting in him electing to have a CRN that was not initially offered. This case highlights that, for selected patients, CRN remains an option. Risk stratification should not be seen as a one-time measurement but should be revisited along the patient's journey as it may impact on treatment decisions.

2. Case presentation

A 57-year-old Caucasian male presented with a 2-month history of a cough, weight loss, chest discomfort and night sweats to the respiratory physicians. Subsequent investigations revealed a 6.5cm right renal tumour with clear evidence of multifocal bilateral pulmonary metastases measuring up to 1cm in diameter [Fig. 1].

The patient had a history of gastro-intestinal reflux disorder (GORD), well controlled with a PPI but was otherwise fit and well with no past surgical history. He worked in the maintenance department of the local airport. He was a non-smoker and a social drinker with a Karnofsky performance status of 90%.

3. Investigations

He underwent a CT guided renal biopsy that confirmed the presence of clear cell renal carcinoma (RCC). Baseline blood investigations showed him to be anaemic with a Haemoglobin of 117 g/L, a normal Neutrophil count of $5.3 \times 10^9/L$, a high platelet count of $427 \times 10^9/L$ and a normal corrected Ca of 2.34 mmol/L making him poor risk according to International Metastatic RCC Database Consortium (IMDC) Criteria.

4. Treatment

He was discussed at the regional renal cancer MDT. In view of the diagnosis of poor risk metastatic clear cell renal cancer he was referred to the oncologists with a view systemic treatment and was being considered for the Cosmic-313 trial (A study of Cabozantinib in Combination With Nivolumab and Ipilimumab in Patients With Previously Untreated Advanced or Metastatic Renal Cell Carcinoma). Prior to being enrolled in this trial he was restaged. A further CT at a 3-month interval showed a significant reduction in the size of his pulmonary metastases and his Full Blood Count normalised with a consequent improvement in his IMDC risk stratification from poor to intermediate risk without

^{*} Corresponding author. 1345 Govan Road, Queen Elizabeth University Hospital, G51 4TF, UK.

E-mail address: kay.mcgillivray@nhs.scot (K. McGillivray).

<https://doi.org/10.1016/j.eucr.2023.102561>

Received 7 September 2023; Accepted 10 September 2023

Available online 30 September 2023

2214-4420/Crown Copyright © 2023 Published by Elsevier Inc.

(<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

This is an open access article under the CC BY-NC-ND license

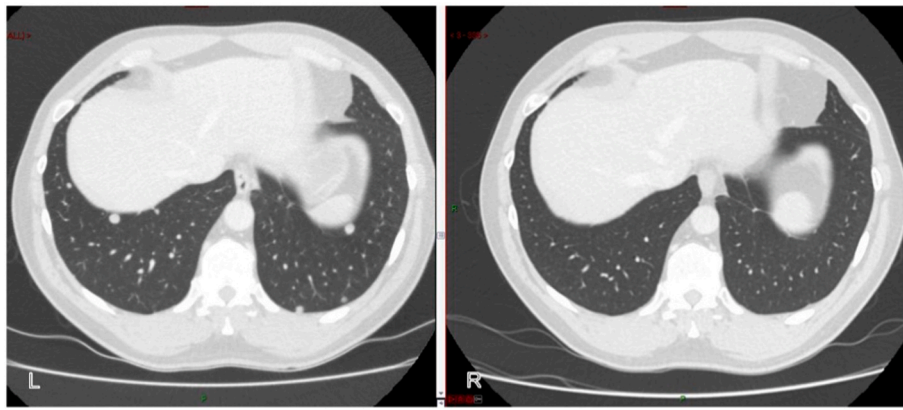


Fig. 1. CT of Thorax. Image on left (L)(July 2020) shows bilateral pulmonary metastases. Image on right (R)(July 2021) shows resolution of these lesions.

treatment. He was re-discussed at the regional renal cancer MDT and subsequently underwent a cytoreductive nephrectomy with the pathology report confirming an ISUP grade 4 pT3a clear cell renal carcinoma and a Leibovich score of 8 that was locally completely excised. He made an uneventful post-operative recovery.

5. Outcome and follow-up

At 12 months follow up his CT continues to show resolution of his pulmonary metastases and no evidence of other recurrent nodal or metastatic disease [Fig. 1]. He has still received no systemic treatment.

6. Discussion

Renal Cancer is the 7th commonest Cancer in the UK accounting for 4% of all malignancies and up to one third of patients will have metastases at diagnosis.

Following the publication of 2 trials, run by the EORTC and the Southwest Oncology Group in the US, early in the millennium, cytoreductive nephrectomy (CRN) became the standard of care for fit patients with metastatic RCC prior to systemic therapy. The combined analysis of these trials reported a 5.8 month survival advantage for patients treated with CRN plus interferon alpha-2b versus interferon alpha-2b alone.¹

With the advent of therapies targeting the vascular endothelial growth factor (VEGF) pathway or the mammalian target of rapamycin pathway (MTOR), the benefits of CRN became less clear. Daniel Heng identified 6 factors from the International Metastatic Renal Cell Carcinoma Database Consortium (IMDC) (haemoglobin less than the lower limit of normal (LLN); corrected calcium greater than the upper limit of normal (ULN); neutrophils greater than the ULN; platelets greater than the ULN; Karnofsky performance status less than 80% and time from diagnosis to treatment of less than 1 year) that were independent predictors of overall survival in metastatic RCC.² He used these factors to stratify patients into good, intermediate or poor risk categories dependent on whether they had 0, 1 to 2 or 3 or more of these indices. He subsequently went on to show that patients with 4 or more of these factors did not benefit at all from CRN and concluded that only patients with good or intermediate risk disease, likely to survive for more than 12 months, should be considered for CRN.³

CRN fell further from favour following the publication of the CAR-MENA trial in 2018 which concluded that Sunitinib alone was not inferior to nephrectomy followed by Sunitinib in patients with Metastatic RCC who were classified as having intermediate or poor risk disease.⁴ We are now moving into an era where immune checkpoint inhibitors are becoming first line treatment for metastatic RCC and a randomised trial looking at CRN with these agents has yet to be performed. The current EAU guidelines only recommend immediate CRN in patients with good performance status who do not require immediate

systemic therapy or in patients where synchronous excision of all oligometastatic disease is possible.⁵ Delayed CRN remains an option for patients who respond well to systemic therapy.

Spontaneous Regression of metastatic clear cell RCC is a rare but well recognised phenomenon occurring in up to 7% of all patients.⁶ It is most often seen in patients with isolated pulmonary metastases but has also been reported for brain, bone adrenal and liver metastases. It is usually seen in patients who have previously undergone a nephrectomy, although here we report spontaneous regression of untreated pulmonary metastatic disease in a patient with clear cell renal cancer with the primary tumour in situ and we believe this to be the first such case in the literature. The concurrent improvement in this patient's risk stratification resulted in a change of treatment recommendation by the MDT. As the patient had metastatic disease and fell into the poor risk category due to his anaemia and high platelet count, immediate systemic treatment was initially advised. With regression of his metastatic disease and normalisation of his Full Blood Count he was reclassified as intermediate risk and underwent a CRN as, with such a low residual disease burden, he did not require immediate systemic treatment.

In the current metastatic RCC treatment paradigm, CRN continues to make way for more effective systemic treatments, such as immune checkpoint inhibitors. As patient fitness decreases and disease volume and the number of IMDC risk factors increase, the benefits of CRN appear to be less. It does, however, continue to have a role and careful patient selection for upfront or deferred CRN, based on risk stratification, remains key.

This case highlights that spontaneous regression of RCC is not simply a radiological phenomenon. The actual cause is unknown but, RCC is unusual amongst solid tumours due to its immunogenic properties.⁷ It is understandable that spontaneous regression could affect systemic markers of disease such as the Full Blood Count. When managing patients with metastatic RCC, it is important to recognise prognostic scores are not a one-time measurement but may alter during the course of treatment. Such changes in risk stratification may have a significant impact on a patient's treatment options. Specifically, in patients with metastatic RCC the option of CRN should be continually revisited over the course of a patient's journey.

7. Learning points

- Spontaneous regression of metastatic disease in RCC can occur
- CRN is no longer advised for all patients with metastatic RCC
- Prognostic stratification of metastatic RCC should not be a one-time measurement and can alter over a patient's journey
- It is important to revisit CRN as a treatment option for patients with metastatic RCC who respond to treatment

References

1. Flanigan RC, Mickisch G, Sylvester R, Tangen C, Van Poppel H, Crawford ED. Cytoreductive nephrectomy in patients with metastatic renal cancer: a combined analysis. *J Urol*. 2004 Mar;171(3):1071–1076.
2. Heng DY, Xie W, Regan MR, et al. Prognostic factors for overall survival in patients with metastatic renal cell carcinoma treated with vascular endothelial growth factor-targeted agents: results from a large, multicenter study. *J Clin Oncol*. 2009;27(34):5794–5799.
3. Heng DY, Connor Wells J, Rini BR, et al. Cytoreductive nephrectomy in patients with synchronous metastases from renal cell carcinoma: results from the international metastatic renal cell carcinoma Database Consortium. *Eur Urol*. 2014;66:704–710.
4. Méjean A, Ravaud A, Thezenas S, et al. Sunitinib alone or after nephrectomy in metastatic renal-cell carcinoma. *N Engl J Med*. 2018 Aug 2;379(5):417–427.
5. Bex A, Albiges L, Ljungberg B, et al. Updated European association of urology guidelines for cytoreductive nephrectomy in patients with synchronous metastatic clear-cell renal cell carcinoma. *Eur Urol*. 2018;74:805–809.
6. Maruschke M, Anastasiadis AG, Hakenberg OW. Spontaneous regression of renal cell carcinoma: reality or myth? *World J Clin Urol*. 2014;3(3):201–208. November 24.
7. Childs R, Chernoff A, Contentin N, et al. Regression of metastatic renal-cell carcinoma after nonmyeloablative allogeneic peripheral-blood stem-cell transplantation. *N Engl J Med*. 2000 Sep 14;343(11):750–758.